Inheritance

Mark scheme and worked solutions · 60 questions

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Part 1

0610_T17_P1_20261018

Part 1 · Question 1

core · 1 mark · 0610_T17_P1_20261018_Q1

Answer: B

Rationale
Chromosomes are made of DNA. Starch and cellulose are carbohydrates, and fat is a lipid; none is the chromosome's genetic material.

Part 1 · Question 2

core · 1 mark · 0610_T17_P1_20261018_Q2

Answer: D

Rationale
A gene is a length of DNA coding for a protein. It is not a cell, membrane or the protein it encodes.

Part 1 · Question 3

core · 1 mark · 0610_T17_P1_20261018_Q3

Answer: A

Rationale
Alleles are alternative forms of one gene. Ribosomes make proteins, gametes are reproductive cells, and enzymes are proteins with catalytic roles.

Part 1 · Question 4

core · 1 mark · 0610_T17_P1_20261018_Q4

Answer: C

Rationale
Human males normally have XY and females XX in body cells. A single X is a gamete complement, not the normal male body-cell pair.

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Part 1 · Question 5

core · 1 mark · 0610_T17_P1_20261018_Q5

Answer: D

Rationale
One X from each gamete gives XX. The sperm in this example contributes no Y and each normal gamete contributes one sex chromosome.

Part 1 · Question 6

supplement · 1 mark · 0610_T17_P1_20261018_Q6

Answer: B

Rationale
The order of DNA bases determines the order of amino acids. Cell position and starch do not specify that sequence.

Part 1 · Question 7

supplement · 1 mark · 0610_T17_P1_20261018_Q7

Answer: C

Rationale
Different amino-acid sequences can produce different folding and shapes. Identical function or active sites is not guaranteed.

Part 1 · Question 8

supplement · 1 mark · 0610_T17_P1_20261018_Q8

Answer: A

Rationale
DNA controls production of proteins including carriers. A changed carrier may alter transport without removing DNA or changing every protein.

Part 1 · Question 9

supplement · 1 mark · 0610_T17_P1_20261018_Q9

Answer: B

Rationale
mRNA copies genetic information and moves to the cytoplasm. Chromosomes normally remain in the nucleus during this process.

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Part 1 · Question 10

supplement · 1 mark · 0610_T17_P1_20261018_Q10

Answer: C

Rationale
Ribosomes assemble amino acids in the order specified by mRNA. The other locations are not the site of protein assembly.

Part 1 · Question 11

supplement · 1 mark · 0610_T17_P1_20261018_Q11

Answer: D

Rationale
Most body cells have the same genes but express different subsets. Specialisation does not normally require loss of all DNA or entirely different genomes.

Part 1 · Question 12

supplement · 1 mark · 0610_T17_P1_20261018_Q12

Answer: A

Rationale
Haploid means one set of chromosomes. Diploid means two sets; neither term describes a protein or a pair of cells.

Part 1 · Question 13

supplement · 1 mark · 0610_T17_P1_20261018_Q13

Answer: C

Rationale
Diploid nuclei contain two chromosome sets. This refers to chromosome sets, not a fixed total of one chromosome or amino acids.

Part 1 · Question 14

supplement · 1 mark · 0610_T17_P1_20261018_Q14

Answer: A

Rationale
There are 23 pairs, giving 46 chromosomes. Twenty-three is the normal haploid number; 44 counts only the autosomes.

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Part 1 · Question 15

supplement · 1 mark · 0610_T17_P1_20261018_Q15

Answer: B

Rationale
The mRNA base sequence specifies the amino-acid sequence. Oxygen availability can affect metabolism but does not encode the sequence.

Part 1 · Question 16

core · 4 marks · 0610_T17_P1_20261018_Q16

(a) [2]

- Chromosomes are made of DNA. (1)
- A gene is a length of DNA that codes for a protein. (1)

(b) [2]

- An allele is an alternative form of a gene. (1)
- Different alleles have differences in their DNA sequence. (1)

Worked solution
(a) Chromosomes are made of DNA. A gene is a length of DNA that codes for a protein.

(b) An allele is an alternative form of a gene. Different alleles have differences in their DNA sequence.

Part 1 · Question 17

core · 4 marks · 0610_T17_P1_20261018_Q17

(a) [2]

- The mother has XX. (1)
- The father has XY. (1)

(b) [2]

- Eggs each contain an X chromosome. (1)
- Sperm contain either X or Y, giving XX or XY after fertilisation. (1)

Worked solution
(a) The mother has XX. The father has XY.

(b) Eggs each contain an X chromosome. Sperm contain either X or Y, giving XX or XY after fertilisation.

Part 1 · Question 18

supplement · 4 marks · 0610_T17_P1_20261018_Q18

(a) [2]

- The DNA base sequence differs between the alleles. (1)
- The base sequence determines the order of amino acids in the protein. (1)

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(b) [2]

- A changed amino-acid sequence can alter the protein's shape. (1)
- A changed active-site shape may reduce binding of the usual substrate. (1)

Worked solution
(a) The DNA base sequence differs between the alleles. The base sequence determines the order of amino acids in the protein.

(b) A changed amino-acid sequence can alter the protein's shape. A changed active-site shape may reduce binding of the usual substrate.

Part 1 · Question 19

supplement · 4 marks · 0610_T17_P1_20261018_Q19

(a) [2]

- The gene controls production of a protein, the enzyme. (1)
- The enzyme catalyses a reaction, so its production affects that reaction. (1)

(b) [2]

- Membrane carrier proteins. (1)
- Receptors for neurotransmitters. (1)

Worked solution
(a) The gene controls production of a protein, the enzyme. The enzyme catalyses a reaction, so its production affects that reaction.

(b) Membrane carrier proteins. Receptors for neurotransmitters.

Part 1 · Question 20

supplement · 4 marks · 0610_T17_P1_20261018_Q20

(a) [2]

- An mRNA copy of the gene is made in the nucleus. (1)
- The mRNA moves from the nucleus to the cytoplasm. (1)

(b) [2]

- The mRNA passes through a ribosome. (1)
- The ribosome assembles amino acids in the sequence specified by the mRNA. (1)

Worked solution
(a) An mRNA copy of the gene is made in the nucleus. The mRNA moves from the nucleus to the cytoplasm.

(b) The mRNA passes through a ribosome. The ribosome assembles amino acids in the sequence specified by the mRNA.

Part 1 · Question 21

supplement · 4 marks · 0610_T17_P1_20261018_Q21

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(a) [2]

- Most body cells in one individual contain the same genes. (1)
- A gene can be present without being expressed. (1)

(b) [2]

- Different cells express different genes. (1)
- They produce the particular proteins needed for their specific functions. (1)

Worked solution
(a) Most body cells in one individual contain the same genes. A gene can be present without being expressed.

(b) Different cells express different genes. They produce the particular proteins needed for their specific functions.

Part 1 · Question 22

supplement · 4 marks · 0610_T17_P1_20261018_Q22

(a) [2]

- Diploid means containing two sets of chromosomes. (1)
- The haploid number is 8. (1)

(b) [2]

- A haploid nucleus contains one set of chromosomes. (1)
- One set is half the diploid number, so 16 divided by 2 is 8. (1)

Worked solution
(a) Diploid means containing two sets of chromosomes. The haploid number is 8.

(b) A haploid nucleus contains one set of chromosomes. One set is half the diploid number, so 16 divided by 2 is 8.

Part 1 · Question 23

supplement · 4 marks · 0610_T17_P1_20261018_Q23

(a) [2]

- A body-cell nucleus contains 46 chromosomes. (1)
- A gamete nucleus contains 23 chromosomes. (1)

(b) [2]

- The body-cell nucleus is diploid with two sets. (1)
- The gamete nucleus is haploid with one set. (1)

Worked solution
(a) A body-cell nucleus contains 46 chromosomes. A gamete nucleus contains 23 chromosomes.

(b) The body-cell nucleus is diploid with two sets. The gamete nucleus is haploid with one set.

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Part 1 · Question 24

supplement · 4 marks · 0610_T17_P1_20261018_Q24

(a) [2]

- The gene remains in the nucleus. (1)
- An mRNA copy moves to the cytoplasm instead. (1)

(b) [2]

- A ribosome assembles amino acids to form the protein. (1)
- The sequence is specified by the mRNA rather than DNA being converted directly into protein. (1)

Worked solution

(a) The gene remains in the nucleus. An mRNA copy moves to the cytoplasm instead.

(b) A ribosome assembles amino acids to form the protein. The sequence is specified by the mRNA rather than DNA being converted directly into protein.

Part 1 · Question 25

supplement · 4 marks · 0610_T17_P1_20261018_Q25

(a) [2]

- DNA controls production of the receptor protein. (1)
- Whether the receptor is present affects the cell's ability to detect the signal. (1)

(b) [2]

- The receptor gene is expressed in one cell type. (1)
- It is not expressed, or is expressed much less, in the other cell type. (1)

Worked solution

(a) DNA controls production of the receptor protein. Whether the receptor is present affects the cell's ability to detect the signal.

(b) The receptor gene is expressed in one cell type. It is not expressed, or is expressed much less, in the other cell type.

Part 1 · Question 26

core · 4 marks · 0610_T17_P1_20261018_Q26

(a) [2]

- Some sperm contain X. (1)
- Other sperm contain Y. (1)

(b) [2]

- Fertilisation by an X-bearing or Y-bearing sperm is a new event. (1)
- Each outcome remains approximately equally likely regardless of earlier children. (1)

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Worked solution
(a) Some sperm contain X. Other sperm contain Y.

(b) Fertilisation by an X-bearing or Y-bearing sperm is a new event. Each outcome remains approximately equally likely regardless of earlier children.

Part 1 · Question 27

supplement · 4 marks · 0610_T17_P1_20261018_Q27

(a) [2]

- The sequence of bases in the gene specifies amino-acid order. (1)
- The two individuals may have different alleles of that gene. (1)

(b) [2]

- Different amino-acid sequences can produce different protein shapes. (1)
- A change in shape may affect binding or transport of the usual substance. (1)

Worked solution
(a) The sequence of bases in the gene specifies amino-acid order. The two individuals may have different alleles of that gene.

(b) Different amino-acid sequences can produce different protein shapes. A change in shape may affect binding or transport of the usual substance.

Part 1 · Question 28

supplement · 4 marks · 0610_T17_P1_20261018_Q28

(a) [2]

- The amino acids could be assembled in different sequences. (1)
- The correct order must be specified by genetic information. (1)

(b) [2]

- The mRNA base sequence supplies the sequence information. (1)
- The ribosome reads that information while assembling the amino acids. (1)

Worked solution
(a) The amino acids could be assembled in different sequences. The correct order must be specified by genetic information.

(b) The mRNA base sequence supplies the sequence information. The ribosome reads that information while assembling the amino acids.

Part 1 · Question 29

supplement · 4 marks · 0610_T17_P1_20261018_Q29

(a) [2]

- P has 10 divided by 2, giving 5. (1)
- Q has 28 divided by 2, giving 14. (1)

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(b) [2]

- Diploid describes two complete sets, regardless of the number in each set. (1)
- Both species have chromosome pairs, although the number of pairs differs. (1)

Worked solution
(a) P has 10 divided by 2, giving 5. Q has 28 divided by 2, giving 14.

(b) Diploid describes two complete sets, regardless of the number in each set. Both species have chromosome pairs, although the number of pairs differs.

Part 1 · Question 30

supplement · 4 marks · 0610_T17_P1_20261018_Q30

(a) [2]

- A chromosome is made of DNA. (1)
- A gene is a length of that DNA coding for a protein. (1)

(b) [2]

- The changed gene can alter production of the enzyme protein. (1)
- Less enzyme may reduce the rate of the metabolic reaction it catalyses. (1)

Worked solution
(a) A chromosome is made of DNA. A gene is a length of that DNA coding for a protein.

(b) The changed gene can alter production of the enzyme protein. Less enzyme may reduce the rate of the metabolic reaction it catalyses.

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Part 2

0610_T17_P2_20261018

Part 2 · Question 1

supplement · 1 mark · 0610_T17_P2_20261018_Q1

Answer: B

Rationale
Mitosis gives genetically identical nuclei and maintains chromosome number. Halving is associated with meiosis and fusion with fertilisation.

Part 2 · Question 2

supplement · 1 mark · 0610_T17_P2_20261018_Q2

Answer: D

Rationale
Exact replication provides copies that can be separated into daughter nuclei. Destruction, fertilisation and reduction division are different processes.

Part 2 · Question 3

supplement · 1 mark · 0610_T17_P2_20261018_Q3

Answer: A

Rationale
Stem cells divide by mitosis and their daughter cells can specialise. They contain chromosomes and are not defined by pollen production.

Part 2 · Question 4

supplement · 1 mark · 0610_T17_P2_20261018_Q4

Answer: C

Rationale
Meiosis produces gametes with reduced chromosome number. Fertilisation joins gametes, while pollination and digestion have other roles.

Part 2 · Question 5

supplement · 1 mark · 0610_T17_P2_20261018_Q5

Answer: D

Rationale
Meiosis halves 32 to 16 and produces genetically different cells. Neither chromosome doubling nor repeated halving to eight describes the process.

Part 2 · Question 6

core · 1 mark · 0610_T17_P2_20261018_Q6

Answer: B

Rationale
Phenotype means observable features. Rr is the genotype; R and r are alleles, and a gamete is a reproductive cell.

Part 2 · Question 7

core · 1 mark · 0610_T17_P2_20261018_Q7

Answer: C

Rationale
Heterozygous means two different alleles of one gene, here T and t. TT and tt are homozygous; the unrelated notation is not a genotype for this gene.

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Part 2 · Question 8

core · 1 mark · 0610_T17_P2_20261018_Q8

Answer: A

Rationale
The recessive feature is expressed when no dominant allele is present, as in bb. BB and Bb contain B; one allele alone is not the stated diploid genotype.

Part 2 · Question 9

core · 1 mark · 0610_T17_P2_20261018_Q9

Answer: B

Rationale
Pp produces P and p gametes; pp produces p gametes only. The offspring are equally likely Pp or pp, giving 1 : 1.

Part 2 · Question 10

core · 1 mark · 0610_T17_P2_20261018_Q10

Answer: C

Rationale
A 3 : 1 ratio under complete dominance includes one homozygous recessive combination among four equally likely combinations. Both other genotypes express the dominant feature.

Part 2 · Question 11

supplement · 1 mark · 0610_T17_P2_20261018_Q11

Answer: D

Rationale
A recessive partner allows recessive offspring to reveal a recessive allele in the unknown parent. A dominant partner can mask that distinction.

Part 2 · Question 12

supplement · 1 mark · 0610_T17_P2_20261018_Q12

Answer: A

Rationale
In codominance both alleles contribute to the heterozygote's phenotype. Neither allele is simply absent or completely masked.

Part 2 · Question 13

supplement · 1 mark · 0610_T17_P2_20261018_Q13

Answer: C

Rationale
Group O requires two I^o alleles. I^A and I^B are each dominant over I^o, and together they give group AB.

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Part 2 · Question 14

supplement · 1 mark · 0610_T17_P2_20261018_Q14

Answer: A

Rationale
A male's single X can express a recessive colour-blindness allele without another X carrying a dominant allele to mask it.

Part 2 · Question 15

core · 1 mark · 0610_T17_P2_20261018_Q15

Answer: B

Rationale
Two aa parents produce only a gametes, so all offspring are aa. Crosses containing heterozygotes can produce more than one genotype.

Part 2 · Question 16

supplement · 4 marks · 0610_T17_P2_20261018_Q16

(a) [2]

- Each daughter nucleus has 24 chromosomes. (1)
- Copies of each replicated chromosome separate so each nucleus receives a complete set. (1)

(b) [2]

- Growth by increasing cell number. (1)
- Asexual reproduction or replacement of worn-out cells. (1)

Worked solution
(a) Each daughter nucleus has 24 chromosomes. Copies of each replicated chromosome separate so each nucleus receives a complete set.

(b) Growth by increasing cell number. Asexual reproduction or replacement of worn-out cells.

Part 2 · Question 17

supplement · 4 marks · 0610_T17_P2_20261018_Q17

(a) [2]

- Each daughter nucleus needs a complete set of genetic information. (1)
- Exact copying helps both daughter nuclei receive the same information as the parent. (1)

(b) [2]

- The copies separate. (1)
- Each daughter nucleus receives one copy of each chromosome. (1)

Worked solution
(a) Each daughter nucleus needs a complete set of genetic information. Exact copying helps both daughter nuclei receive the same information as the parent.

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(b) The copies separate. Each daughter nucleus receives one copy of each chromosome.

Part 2 · Question 18

supplement · 4 marks · 0610_T17_P2_20261018_Q18

(a) [2]

- Stem cells are unspecialised cells. (1)
- They divide by mitosis to produce daughter cells. (1)

(b) [2]

- The daughter cells become specialised. (1)
- They develop the features needed for that tissue's particular function. (1)

Worked solution

(a) Stem cells are unspecialised cells. They divide by mitosis to produce daughter cells.

(b) The daughter cells become specialised. They develop the features needed for that tissue's particular function.

Part 2 · Question 19

supplement · 4 marks · 0610_T17_P2_20261018_Q19

(a) [2]

- Each gamete nucleus has 6 chromosomes. (1)
- Gametes are genetically different from one another. (1)

(b) [2]

- It halves the number of chromosome sets. (1)
- It changes a diploid nucleus to haploid nuclei. (1)

Worked solution

(a) Each gamete nucleus has 6 chromosomes. Gametes are genetically different from one another.

(b) It halves the number of chromosome sets. It changes a diploid nucleus to haploid nuclei.

Part 2 · Question 20

core · 4 marks · 0610_T17_P2_20261018_Q20

(a) [2]

- Hh is its genotype, describing the alleles present. (1)
- Hairy leaves are its phenotype, an observable feature. (1)

(b) [2]

- Inheritance is transmission of genetic information from generation to generation. (1)
- Alleles of genes are passed from parents to offspring. (1)

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Worked solution
(a) Hh is its genotype, describing the alleles present. Hairy leaves are its phenotype, an observable feature.

(b) Inheritance is transmission of genetic information from generation to generation. Alleles of genes are passed from parents to offspring.

Part 2 · Question 21

core · 4 marks · 0610_T17_P2_20261018_Q21

(a) [2]

- BB and bb are homozygous because each has two identical alleles. (1)
- Bb is heterozygous because its alleles differ. (1)

(b) [2]

- Bb is black because the dominant B allele is expressed. (1)
- bb is white because no dominant B allele is present. (1)

Worked solution
(a) BB and bb are homozygous because each has two identical alleles. Bb is heterozygous because its alleles differ.

(b) Bb is black because the dominant B allele is expressed. bb is white because no dominant B allele is present.

Part 2 · Question 22

core · 4 marks · 0610_T17_P2_20261018_Q22

(a) [2]

- Each parent produces S and s gametes. (1)
- The offspring combinations are SS, Ss, Ss and ss. (1)

(b) [2]

- The ratio is 3 smooth : 1 wrinkled. (1)
- Only ss offspring are wrinkled because the dominant S allele is absent. (1)

Worked solution
(a) Each parent produces S and s gametes. The offspring combinations are SS, Ss, Ss and ss.

(b) The ratio is 3 smooth : 1 wrinkled. Only ss offspring are wrinkled because the dominant S allele is absent.

Part 2 · Question 23

core · 4 marks · 0610_T17_P2_20261018_Q23

(a) [2]

- The Ll parent produces L and l gametes. (1)
- The ll parent produces l gametes only. (1)

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(b) [2]

- Offspring are equally likely Ll and ll. (1)
- The ratio is 1 long-winged : 1 short-winged. (1)

Worked solution
(a) The Ll parent produces L and l gametes. The ll parent produces l gametes only.

(b) Offspring are equally likely Ll and ll. The ratio is 1 long-winged : 1 short-winged.

Part 2 · Question 24

supplement · 4 marks · 0610_T17_P2_20261018_Q24

(a) [2]

- Cross the red-flowered plant with a white-flowered rr plant. (1)
- Record the flower colours of many offspring. (1)

(b) [2]

- White offspring show the unknown parent supplied r and is Rr. (1)
- A small sample could by chance contain only red offspring even if the parent is Rr. (1)

Worked solution
(a) Cross the red-flowered plant with a white-flowered rr plant. Record the flower colours of many offspring.

(b) White offspring show the unknown parent supplied r and is Rr. A small sample could by chance contain only red offspring even if the parent is Rr.

Part 2 · Question 25

supplement · 4 marks · 0610_T17_P2_20261018_Q25

(a) [2]

- Both parents produce C^R and C^W gametes. (1)
- Offspring genotypes are C^RC^R, C^RC^W and C^WC^W. (1)

(b) [2]

- The ratio is 1 red : 2 mixed red-and-white : 1 white. (1)
- Both alleles contribute to the heterozygote's phenotype because they are codominant. (1)

Worked solution
(a) Both parents produce C^R and C^W gametes. Offspring genotypes are C^RC^R, C^RC^W and C^WC^W.

(b) The ratio is 1 red : 2 mixed red-and-white : 1 white. Both alleles contribute to the heterozygote's phenotype because they are codominant.

Part 2 · Question 26

supplement · 4 marks · 0610_T17_P2_20261018_Q26

(a) [2]

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- The first produces I^A and I^o gametes. (1)
- The second produces I^B and I^o gametes. (1)

(b) [2]

- The combinations give AB, A, B and O. (1)
- These groups occur in an expected ratio of 1 : 1 : 1 : 1. (1)

Worked solution
(a) The first produces I^A and I^o gametes. The second produces I^B and I^o gametes.

(b) The combinations give AB, A, B and O. These groups occur in an expected ratio of 1 : 1 : 1 : 1.

Part 2 · Question 27

supplement · 4 marks · 0610_T17_P2_20261018_Q27

(a) [2]

- X^NX^N. (1)
- X^NX^n. (1)

(b) [2]

- Half the sons are expected to be X^nY. (1)
- A son receives Y from his father and has an equal chance of receiving either maternal X. (1)

Worked solution
(a) X^NX^N. X^NX^n.

(b) Half the sons are expected to be X^nY. A son receives Y from his father and has an equal chance of receiving either maternal X.

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Part 2 · Question 28

core · 4 marks · 0610_T17_P2_20261018_Q28

(a) [2]

- PP is consistent with the line. (1)
- Two PP parents pass on P only, so all offspring are PP and purple. (1)

(b) [2]

- Both parents can pass on p. (1)
- Some offspring will be pp and have white flowers. (1)

Worked solution
(a) PP is consistent with the line. Two PP parents pass on P only, so all offspring are PP and purple.

(b) Both parents can pass on p. Some offspring will be pp and have white flowers.

Part 2 · Question 29

supplement · 4 marks · 0610_T17_P2_20261018_Q29

(a) [2]

- The AB parent is I^AI^B and produces I^A or I^B gametes. (1)
- The O parent is I^oI^o and produces I^o gametes only. (1)

(b) [2]

- Half are expected to be A and half B. (1)
- Each child receives I^o from the O parent, so no child receives both I^A and I^B. (1)

Worked solution

(a) The AB parent is I^AI^B and produces I^A or I^B gametes. The O parent is I^oI^o and produces I^o gametes only.

(b) Half are expected to be A and half B. Each child receives I^o from the O parent, so no child receives both I^A and I^B.

Part 2 · Question 30

core · 4 marks · 0610_T17_P2_20261018_Q30

(a) [2]

- A heterozygote cross predicts three quarters dominant, so 200 × 3 / 4 = 150. (1)
- One quarter recessive gives 200 × 1 / 4 = 50. (1)

(b) [2]

- Which gametes fuse is a chance event. (1)
- An expected probability does not guarantee an exact ratio in a finite sample. (1)

Worked solution

(a) A heterozygote cross predicts three quarters dominant, so 200 × 3 / 4 = 150. One quarter recessive gives 200 × 1 / 4 = 50.

(b) Which gametes fuse is a chance event. An expected probability does not guarantee an exact ratio in a finite sample.

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